

WASP-12b

R-0249

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241128 · created 2026-07-17T06:18:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460642.7524 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.045
Transit depth (Rp/Rs)^2	8.5e-06 +/- 0.0026 [%]
Orbital Inclination (inc)	81.11 +/- 2.89
Airmass coefficient 1 (a1)	0.86 +/- 0.6
Airmass coefficient 2 (a2)	0.28 +/- 0.21
Scatter in the residuals of the lightcurve fit is	78.76 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	7.0
Transit Duration (day)	0.0934 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.045 vs 0.1178 (deviation 1.01σ)
Duration fit vs published	0.0934 d vs 0.125 d (deviation 2.21σ)
Mid-time O–C	-2.1 min
Depth SNR	0.0
Residual scatter	78.76 %

Light curve

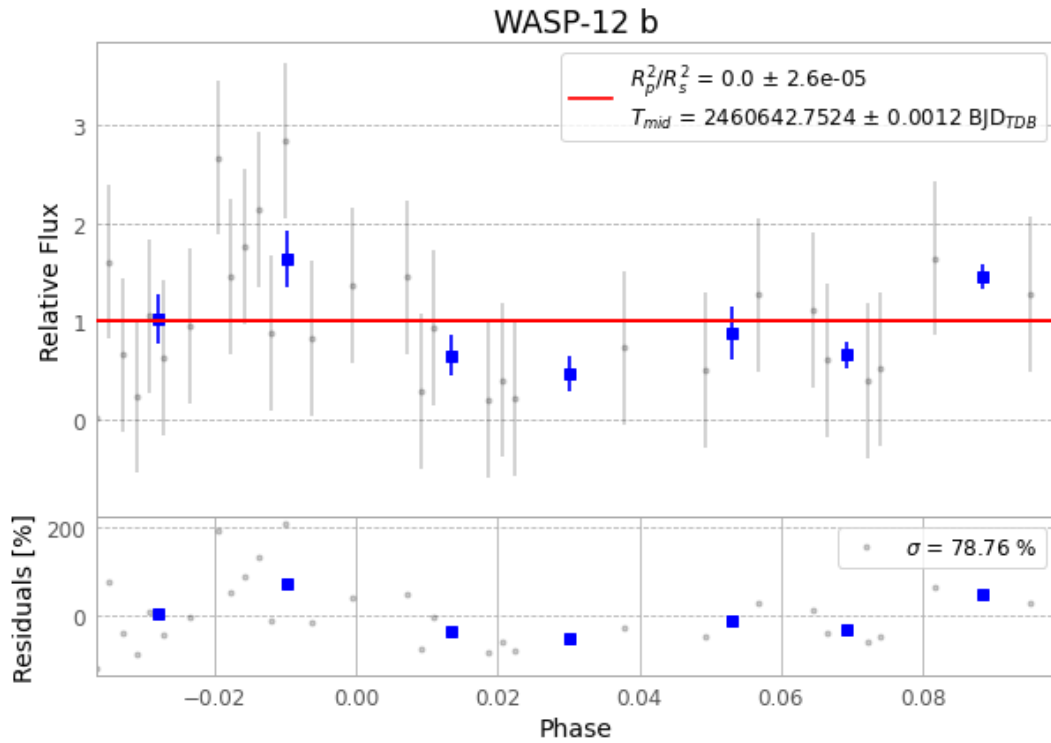


Figure 1 — FinalLightCurve_WASP-12 b_2024-11-27.png (SHA-256 bc128f9a3d0a5df9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [28, 145], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2439952cdc3b6cad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), WASP-12241128045717.FITS → WASP-12241128083315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241128.log (70ba6dcd521e...), FinalLightCurve_WASP-12 b_2024-11-27.png (bc128f9a3d0a...), FinalLightCurve_WASP-12 b_2024-11-27.csv (9a9bbda150bf...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:18Z.